SSS2

Chemistry

Scheme Of Work

The Unified Scheme of Work for Senior Secondary School 2 is designed to help students meet learning objectives, provide comprehensive class notes, and ensure students receive the education they deserve

Includes:

- Schemes of Work
- Recommended Textbooks
- Academic Standards

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The SSS 2 scheme of work was sourced from the Lagos State Ministry of Education: Unified Scheme of Work for Primary Schools

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ABOUT US

Syllabus NG is the premier destination for all educational resources and exam preparation materials in Nigeria and the world. Our mission is simple yet impactful; to empower learners of all ages and backgrounds with the resources they need to succeed academically. It is always better to work smarter than to work harder.

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About SSS2 Scheme of Work

The SS2 unified scheme of work reviewed by the Lagos State Government details the curriculum for students in their second year of senior secondary education. It builds on the foundation from SS1, preparing students for their final year and major external exams like the WASSCE and NECO. Covering core subjects such as English and Mathematics, along with various science, arts, and vocational subjects, the curriculum focuses on both theoretical knowledge and practical skills. It aims to develop well-rounded students with critical thinking abilities and real-world problem-solving skills. Regular reviews ensure the curriculum stays up-to-date with educational standards and societal needs, fostering the development of competent individuals ready for higher education or the workforce.

The primary goals of the SSS1 scheme of work include:

- Providing students with the knowledge and skills needed for academic achievement and personal growth.
- Deepening their understanding of core subjects to prepare them for more specialized studies in SSS2 and SSS3.
- Enhancing their critical thinking, problem-solving skills, and ability to learn independently.
- Promoting ethical behavior, civic responsibility, and a dedication to lifelong learning.
- Preparing students for external exams such as the West African Senior School Certificate Examination (WASSCE) and the National Examination Council (NECO) exams.

To boost student engagement and learning outcomes, the SSS2 scheme of work utilizes various teaching methods, such as:

- Encouraging active participation through discussions, group work, and presentations.
- Including laboratory experiments, field trips, and hands-on activities to support theoretical learning.
- Using digital tools and resources to facilitate learning and provide access to information.
- Involving students in projects that demand critical thinking, creativity, and collaboration.

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Senior Secondary School (SSS 1 - 3)

Lesson Notes For Educators

- Includes Comprehensive Notes
- Easy-to understand
- Curriculum Standard, Class Practice & Assessments
- Arranged Weekly for Each Term

Please send us a message on WhatsApp if you need lesson notes for SSS1-3
(1st, 2nd, and 3rd Term)

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Subjects Available

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- Mathematics
- Biology
- Civic Education
- Agricultural Science
- Chemistry
- Further Mathematics
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- Financial Accounting
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- Book Keeping
- Economics
- French
- Geography
- Government
- Literature
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Introduction

In SS2, students explore chemistry in greater depth. This study year builds upon fundamental knowledge acquired previously, aiming to broaden understanding and practical application.

In the Lagos state unified scheme of work for chemistry in SS2, topics typically cover areas such as chemical bonding, basic organic chemistry, and stoichiometry, as well as more advanced concepts like equilibrium and reaction kinetics. Practical sessions often involve experiments to reinforce theoretical learning, encouraging a hands-on approach to comprehending chemical principles.

SS2 chemistry is a foundational stepping stone for future studies and careers in fields such as medicine, engineering, and environmental science, emphasizing the development of critical thinking and problem-solving skills crucial for scientific exploration. By the end of SS2, students will have a strong foundation in Chemistry, preparing them for more advanced topics in the future.

Achievement standards
SSS2 First Term
At the end of SSS2 first term, students can;

- Classify blocks of elements in the periodic table into families;
- Outline the redox reactions in nature namely; Photosynthesis, Respiration, Combustion, Corrosion, Rust.
- Explain the mechanism of electrolysis and apply it to daily life.

SSS2 Second Term
At the end of SSS2 second term, students can;

- Use different renewable energy sources to power light in schools without using light from PHCN;
- Recycle waste materials e.g. plastic bottles, cardboard boxes, packaging materials, household and agricultural waste,

polyethylene waste e.g. sachet of water plastic bags, crown metal cover, metal scraps, old clothes, etc;
- Prepare antiseptics and ointments using petroleum jelly and grounded sulphur, shea butter and grounded sulphur, and caustic soda.

SSS2 Third Term

At the end of SSS2 third term, students can;

- Titrate solutions of acids against solutions of bases;
- Calculate unknown quantities in Acid-base reactions, Redox titrations, Concentrations in g/dm3 and mol/dm3, % purity and impurity, water crystallization;
- Illustrate hydrogen bonding in water;
- Enumerate the advantages and disadvantages of hard and soft water;
- Prepare saturated, unsaturated, and supersaturated solutions;
- Identify apparatus/uses for volumetric analysis in the laboratory;
- Apply uses of saturated and unsaturated hydrocarbons to daily life.

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Chapter Two

Scheme of Work

First Term

Second Term

Third Term

LAGOS STATE MINISTRY OF EDUCATION UNIFIED SCHEMES OF WORK FOR SENIOR SECONDARY SCHOOLS		
Chemistry Scheme of Work for Senior Secondary Schools 2(SSS2)		
	CLASS	**SS2**
	SUBJECT	**CHEMISTRY**
	TERM	**First Term**
WEEK	**TOPICS**	**Learning Objectives**
1	**PERIODIC TABLE** **- History of periodic table** **- Periodic law** **- Books of elementals s, p,d,f**	At the end of the lesson, students should be able to: i. discuss the history of the periodic table ii. state the periodic law iii. differentiate between a group and a period of elements iv. list the 8 groups and 7 periods of elements with 2 examples of elements in each classification v. specify and name the 4 blocks of elements and give 3 examples in each block of the periodic table vi. classify a given list of elements into s, p, d, f blocks
2	**FAMILIES OF ELEMENTS** **- Alkali metals, alkaline earth metals, noble (rare) gases, halogens, transition metals, etc** **- Characteristics and reactions of each family of elements** **- Changes of periodic properties of elements across the period and down the group (atomic/ionic radius, and volume), ionization energy, electron affinity, electronegativity, etc** **- Diagonal relationships**	At the end of the lesson, students should be able to: i. distinguish the families of elements ii. explain the characteristics and reactions of each family iii. enumerate at least 5 periodic properties of elements iv. discuss the changes/variation of periodic properties of elements across the period and down the group v. explain diagonal relationships in the periodic table

3	**FAMILIES OF ELEMENTS** **- Alkali metals, alkaline earth metals, noble (rare) gases, halogens, transition metals, etc** **- Characteristics and reactions of each family of elements** **- Changes of periodic properties of elements across the period and down the group (atomic/ionic radius, and volume), ionization energy, electron affinity, electronegativity, etc** **- Diagonal relationships**	At the end of the lesson, students should be able to: i. distinguish the families of elements ii. explain the characteristics and reactions of each family iii. enumerate at least 5 periodic properties of elements iv. discuss the changes/variation of periodic properties of elements across the period and down the group v. explain diagonal relationships in the periodic table
4	**OXIDATION AND REDUCTION REACTIONS (REDOX)** **- Definitions of oxidation and reduction** **- Oxidation and reduction in daily life and nature** **- Examples of REDOX equations** **- Reducing and oxidizing agents** **- Rules governing calculation of oxidation number** **- Calculation of oxidation number of elements**	At the end of the lesson, students should be able to: i. identify key words in the topic and sub topics of the lesson ii. explain the meaning of the keywords and relate such to the lesson iii. define oxidation and reduction in 4 different ways iv. mention at least 3 oxidation and reduction processes in daily life and nature v. give examples of REDOX equations vi. define oxidizing and reducing agents vii. identify oxidizing and reducing agents viii. outline rules governing calculation of oxidation numbers ix. calculate the oxidation number of elements in compounds and radicals

5	**OXIDATION AND REDUCTION II** **- IUPAC nomenclature of specific compounds** **- Balancing redox equations in alkaline and acidic media**	At the end of the lesson, students should be able to: i. give IUPAC names of at least 10 specific compounds ii. balance redox equations in alkaline and acidic media
6	**ELECTROLYSIS** **- Definition of terms: electrolysis, electrolyte (strong and weak), non-electrolytes, conductors, non-conductors, electrodes (cathodes and anodes), electrolytic cell** **- Electrovalent and covalent compounds** **- The ionic theory** **- Mechanism of electrolysis**	At the end of the lesson, students should be able to: i. highlight keywords in the topic and sub-topic of the lesson eg electrolysis, electrolyte, electrode, etc ii. articulate the meaning of keywords and relate such to the lesson iii. classify given compounds as strong, weak and non-electrolytes iv. differentiate between electrovalent (ionic) and covalent compounds v. explain the ionic theory vi. explain the mechanism of electrolysis
7	**MID-TERM BREAK**	
8	**ELECTROLYSIS II** **- Factors affecting preferential discharge of ions** **- Faraday's 1st and 2nd laws of electrolysis** **- Electrolysis of specified electrolytes** **- Uses/application of electrolysis (including electroplating)**	At the end of the lesson, students should be able to: i. outline and explain 3 factors affecting preferential discharge of ions during electrolysis ii. state Faraday's 1st and 2nd laws of electrolysis iii. distinguish between Faraday, Columb and Ampere iv. describe and perform experiments on the electrolysis of specified electrolytes v. outline at least 4 uses of electrolysis vi. explain electroplating process

9	**ELECTROLYSIS III** **- Calculations on electrolysis** **- Experiments on electroplating**	At the end of the lesson, students should be able to: i. recall Faraday's 1st and 2nd laws ii. calculate parameters in electrolysis (eg determine the amount of substances liberated at the electrodes during electrolysis, quantity of electricity used, etc) iii. perform experiments on electroplating
10	**ELECTROCHEMICAL CELLS** **- Definition of electrochemical cells** **- Differences between an electrochemical cell and an electrolytic cell** **- Standard electrode potential** **- Electrochemical cell diagram**	At the end of the lesson, students should be able to: i. write key words and their meaning eg electrochemical cells, electrode potential, etc ii. differentiate between an electrolytic cell and an electrochemical cell iii. explain electrode potential iv. define standard electrode potential v. list 2 factors affecting electrode potential of a metal vi. draw a metal/metal ion system vii. write half cell equation viii. draw and label an electrochemical cell
11	**ELECTROCHEMICAL CELL II** **- Types and examples of electrochemical cell** **- E.M.F of a cell** **- Calculations of E.M.F**	At the end of the lesson, students should be able to: i. mention 5 types of electrochemical cells ii. outline 3 examples of each type of electrochemical cell especially in daily life iii. define E.M.F of a cell iv. explicate the relevance of E.M.F of a cell v. calculate the E.M.F of a cell
12	**REVISION/EXAMINATION**	
	CLASS	**SS2**
	SUBJECT	**CHEMISTRY**

	TERM	SECOND TERM
WEEKS	**TOPICS**	**LEARNING OBJECTIVES**
1	**CHEMICAL REACTIONS** **- Basic concepts of reactants and products** **- Rate of reaction** **- Collision theory** **- Activation energy**	At the end of the lesson, students should be able to: i. write balanced equations of reactions ii. indicate reactants and products in chemical equation iii. express/explain the meaning of chemical equations iv. define chemical equations v. explain collision theory vi. explain activation energy
2	**CHEMICAL REACTIONS II** **- Types of chemical reactions** **- Factors affecting the rate of chemical reactions** **- Calculation of rates of chemical reactions**	At the end of the lesson, students should be able to: i. enumerate 5 examples of chemical reactions ii. list and explain at least 5 factors affecting the rate of chemical reactions iii. calculate rates of chemical reactions

3	**ENERGY, ENERGY CHANGES, BIOTECHNOLOGY** **A. RENEWABLE, NON-RENEWABLE ENERGY/BIOTECHNOLOGY** **- Definition and forms of energy** **- Forms and types of energy resources (renewable and non-renewable)** **- Sources of renewable and non-renewable energy** **- Advantages and disadvantages of renewable and non-renewable energy** **B. BIOTECHNOLOGY** **- Projects on renewable and non-renewable energy** **C. EXOTHERMIC AND ENDOTHERMIC REACTIONS** **- Exothermic and endothermic reactions with examples** **- Enthalpy and enthalpy changes** **- Laws of thermodynamics** **- Heat of reactions**	At the end of the lesson, students should be able to: i. identify keywords in the topics and sub topics of the lesson eg renewable, non-renewable, etc ii. explain the meaning of the keywords and relate such to the lesson iii. list forms of energy iv. mention types and forms of energy resources v. differentiate between renewable and non-renewable energy vi. list 5 sources each of renewable and non-renewable energy vii. list 5 advantages each of renewable and non-renewable energy viii. participate in projects on Biotechnology using either renewable or non-renewable energy ix. differentiate between exothermic and endothermic reactions x. draw energy profiles for exothermic and endothermic reactions (catalyzed and un-catalyzed) xi. explain the concept of thermodynamics xii. state the 1st and 2nd laws of thermodynamics xiv. explain enthalpy xv. define various heat of reactions

4	**ENERGY, ENERGY CHANGES AND BIOTECHNOLOGY II** **A. ENTROPY, GIBBS FREE ENERGY** **- Definition of Entropy, Gibbs free energy** **- ΔG, ΔS, ΔH** **- Relationship between ΔG, ΔS, ΔH** **- Conditions for spontaneity in chemical reactions** **- Calculation of ΔG, ΔS, ΔH**	At the end of the lesson, students should be able to: i. write out key words in the topics and sub-topics of the lesson ii. explain ΔG, ΔS, ΔH and the connection between them iii. list the conditions for spontaneity in chemical reactions iv. calculate ΔG, ΔS, ΔH
5	**CHEMICAL EQUILIBRIUM** **- Reversible and irreversible reactions** **- Equlibrium in reversible reactions** **- Equilibrium in nature** **- Le Chatelier's principles** **- Factors affecting equilibrium position in a chemical reaction** **- Expression and calculation of equilibrium constant k**	At the end of the lesson, students should be able to: i. identify keywords in the topic and sub-topic of the lesson ii. express the meaning of the keywords and relate such to the lesson iii. differentiate reversible and irreversible reactions iv. explain the concept of equilibrium in a reversible reaction v. give 5 examples of equilibrium in nature vi. state Le Chatelier's principles and laws of mass action vii. list 3 factors affecting equlibrium position in a chemical reaction viii. write expressions for equilibrium constant K ix. calculate equilibrium constant

6	**NON METALS-HYDROGEN** **- Unique position of Hydrogen in the periodic table** **- Electronic configuration of hydrogen** **- Isotopes of Hydrogen** **- Hydrogen molecule** **- Properties and uses** **- Test for Hydrogen**	At the end of the lesson, students should be able to: i. explain the unique position of Hydrogen on the periodic table ii. list the 3 isotopes of hydrogen iii. write the electronic configuration of hydrogen and it's 3 isotopes iv. show the formation of hydrogen molecule from its electronic structure v. demonstrate the laboratory preparation of hydrogen vi. describe the industrial preparation of hydrogen vii. outline 3 physical and chemical properties of hydrogen viii. list 6 uses of hydrogen ix. test for hydrogen in the laboratory
7	**MID-TERM BREAK**	

8	**NON METALS - OXYGEN** **A. OXYGEN IN NATURE** **- Position of Oxygen on the periodic table** **- Electronic configuration** **- The Oxygen molecule** **- Allotrope of Oxygen** **- Properties and laboratory preparation of Oxygen** **B. OXIDES** **- Definition of oxides** **- Types of oxides (acidic, basic, neutral, amphoteric, higher)** **C. AIR** **- Constituents and it's composition** **- Properties** **- Air pollution and pollutants** **- Effects and control of air pollution** **- Global warming (ozone layer and greenhouse effect)** **D. FLAME** **- Types of flame** **- Zones in flame**	At the end of the lesson, students should be able to: i. discuss oxygen as a component of many natural and chemical substances ii. indicate the position of oxygen on the periodic table iii. write the electronic configuration of oxygen iv. explain the formation of oxygen molecules from it's atomic structure v. mention the only allotrope of oxygen vi. prepare oxygen in the lab vii. describe industrial preparation of oxygen viii. write 4 physical and chemical properties of oxygen ix. outline 5 uses of oxygen x. define oxides xi. list and define at least 4 types of oxides with 1 example each xii. explain why air is a mixture xiii. draw a schematic diagram to show the constituents of air and % composition xiv. enumerate 4 properties of air xv. define air pollution xvi. outline 5 air pollutants xvii. state 5 effects of air pollution xviii. explain 5 methods to combat air pollution xix. discuss global warming in relation to ozone layer and greenhouse effect xx. list 3 types of flames xxi. draw and label a candle + bunsen burner

9	**HALOGENS** **- Meaning, occurrence of halogens in nature and daily life** **- Position of halogens in the periodic table** **- Electronic configuration of halogens** **- Gradations of general properties of halogens (Physical and chemical) down the group** **- Examples of halogen compounds** **- General uses of halogens and their compounds** **- Laboratory preparation of Chlorine** **- Test for halides- Cl, Br, etc** **- Bleaching action of Chlorine**	At the end of the lesson, students should be able to: i. identify key words in the topic and sub-topic of the lesson ii. express the meaning of the keywords and relate such to the lesson iii. briefly discuss occurrence of halogens in nature eg sea water iv. mention 5 substances used in daily life that contains specific halogens eg toothpaste, bleach, etc v. indicate the position of halogens on the periodic table vi. write the electronic configuration of the first four halogens (F, Cl, Br, I) vii. explain the gradation of the physical and chemical properties down the group viii. write and name 5 halogen compounds ix. enumerate 5 general uses of halogens and their compounds x. perform experiment on laboratory preparation of Chlorine gas (if a fume cupboard is available) xi. perform experiments to test for halides in the laboratory xii. demonstrate the bleaching action of Chlorine

10	**NITROGEN AND IT'S COMPOUNDS** **A. NITROGEN** **- Nitrogen in nature and position in the periodic table** **- Electronic configuration of Nitrogen** **- General properties of VA** **- Laboratory preparation of Nitrogen** **- Industrial preparation of Nitrogen from liquid air** **- Properties of Nitrogen (Physical and chemical)** **- Uses of Nitrogen** **- The Nitrogen cycle** **B. OXIDES OF NITROGEN** **- List the oxides of nitrogen (names and formulas) and state their characteristics** **- Explain experiments to distinguish between the oxides** **C. HYDRIDE OF NITROGEN - AMMONIA GAS** **- Chemical formula, physical and chemical properties** **- Uses of ammonia** **- Laboratory preparation of ammonia gas** **- Industrial preparation of ammonia gas (Haber process)** **D. OTHER COMPOUNDS OF NITROGEN**	At the end of the lesson, students should be able to: i. discuss the occurrence of nitrogen in air, nature and food substances ii. indicate the position of nitrogen on the periodic table iii. write the electronic configuration of nitrogen iv. state 3 general properties of VA elements v. explain the laboratory preparation of Nitrogen vi. explain the industrial preparation of Nitrogen vii. list 3 physical and 2 chemical properties of nitrogen viii. list 3 uses of nitrogen ix. draw and explain the nitrogen cycle x. list the 3 main oxides of nitrogen xi. compare the properties of the oxides of nitrogen using a T-Chart xii. explain experiments to distinguish between the 3 oxides of nitrogen xiii. mention ammonia gas as the only hydride of nitrogen and write it's chemical formula xiv. outline 3 properties (physical and chemical) of ammonia xv. explain/demonstrate the laboratory preparation of ammonia gas xvi. explain the industrial preparation of ammonia gas (Haber process) xvii. list 4 uses of ammonia gas xviii. list (names and formulas) of 5 other compounds of nitrogen in the laboratory eg Trioxonitrate (v), ammonium salts, etc xix. list 3 uses of Trioxonitrate (v) and ammonium salts xx. perform experiments to test for Trioxonitrate (v) ion xxi. perform experiments to test for NH4

11	**A. SULPHUR** **- Sulphur in nature and position on the periodic table** **- Electronic configuration of Sulphur** **- General properties of VA** **- Explain the concept of allotropy** **- List the allotropes of sulphur** **- Explain the extraction of sulphur (French process)** **- State the uses of sulphur** **- Oxidation number of Sulphur in it's major compound** **B. COMPOUNDS OF SULPHUR** **- H2S** **- H2SO3** **- H2SO4** **- Oxides of sulphur** **- Industrial preparation of H2SO4** **- Uses of H2SO4**	At the end of the lesson, students should be able to: i. mention the local name of sulphur in their language ii. write 5 substances used in daily life and nature that contain sulphur eg antiseptic soap iii. indicate the position of sulphur on the periodic table iv. write the electronic configuration of sulphur v. enumerate 3 general properties of sulphur vi. recall the concept of allotropy treated under carbon vii. explain the concept of allotropy viii. describe at least 3 isotopes of sulphur ix. explain the extraction of sulphur x. enumerate 5 uses of sulphur xi. calculate the oxidation scores of sulphur in its main compounds using rules taught under oxidation and reduction xii. list a few major compounds of sulphur xiii. list their characteristic properties xiv. explain the industrial preparation of H2SO4 xv. perform experiment test
12	**REVISION/EXAMINATIONS**	
	CLASS	**SS2**
	SUBJECT	**CHEMISTRY**
	TERM	**THIRD TERM**
WEEKS	**TOPICS**	**LEARNING OBJECTIVES**

1	**WATER** **- Structure of water** **- Laboratory preparation of water** **- Properties of water (Physical and chemical)** **- Types of water (Soft and hard)** **- Types of hardness (Temporary and Permanent)** **- Causes of hardness** **- Advantages and disadvantages of hard water** **- Methods of removal of hardness/ softening water** **- Purification of water from town (municipal) supply**	At the end of the lesson, students should be able to: i. discuss water as an essential substance for life and major components of living tissue ii. mention elemental component of water iii. draw the structure of water iv. explain the laboratory preparation of water v. list 5 physical properties of water and explain the unusual high boiling of water vi. outline at least 3 chemical properties of water viii. mention the types and causes of hardness in water ix. discuss at least 3 advantages and disadvantages of soft and hard water x. outline and describe at least 4 methods of softening water/removal of hardness xi. outline and explain 5 processes involved in the treatment of water for town/municipal supply

2	**WATER** **- Structure of water** **- Laboratory preparation of water** **- Properties of water (Physical and chemical)** **- Types of water (Soft and hard)** **- Types of hardness (Temporary and Permanent)** **- Causes of hardness** **- Advantages and disadvantages of hard water** **- Methods of removal of hardness/ softening water** **- Purification of water from town (municipal) supply**	At the end of the lesson, students should be able to: i. discuss water as an essential substance for life and major components of living tissue ii. mention elemental component of water iii. draw the structure of water iv. explain the laboratory preparation of water v. list 5 physical properties of water and explain the unusual high boiling of water vi. outline at least 3 chemical properties of water viii. mention the types and causes of hardness in water ix. discuss at least 3 advantages and disadvantages of soft and hard water x. outline and describe at least 4 methods of softening water/removal of hardness xi. outline and explain 5 processes involved in the treatment of water for town/municipal supply

3	**SOLUBILITY** **- Definition of terms: solute, solvent, solution, saturated, un-saturated and super saturated solution, solubility)** **- Factors affecting solubility** **- Solubility curve** **- Calculations on solubility** **- Applications of solubility and solubility curve**	At the end of the lesson, students should be able to: i. identify keywords in the topic and sub-topic of the lesson ii. explain the meaning of the keywords and relate such to the lesson iii. give 5 examples each of solute, solvent and solution iv. prepare solution = solute + solvent v. prepare saturated, un-saturated, super saturated solution vi. differentiate between saturated, un-saturated and super saturated solutions vii. define solubility viii. list 3 factors affecting solubility ix. calculate the solubility of solvents and solutes x. plot solubility curves xi. list and discuss at least 3 applications of solubility and solubility curve in daily life and nature
4	**MASS-VOLUME RELATIONSHIP** **- Mole concept, molar quantities** **- Relative densities, relative molar mass, molar volume, S.T.P, S. V.P** **- Molar concentration and mass concentration** **- Chemical equations** **- Calculations involving mass and volume**	At the end of the lesson, students should be able to: i. identify keywords in the topic and sub-topic of the lesson ii. express the meaning of the keywords and relate such to the lesson iii. explain concept of mole, molar volume, molar mass, molar concentration, etc iv. calculate the mass/volume of reactants and products, number of moles of reacting substances and products, relative densities v. derive S.I units of calculative quantities

5	**VOLUMETRIC/QUANTITATIVE ANALYSIS** **- Definition of terms: titration, trimetry, molar solution, standard solution, indicator, etc** **- Apparatus required for volumetric analysis** **- Precautions in titration** **- Steps in titration** **- Recording acceptable titre value** **- Acid-base and redox titrations** **- Determination of:** **a) concentration in g/dm and mol/dm** **b) % purity and impurity** **c) water of crystallization** **- Heat of neutralization**	At the end of the lesson, students should be able to: i. identify topics and sub-topics of the lesson ii. express the meaning of the keywords and relate such to the lesson iii. explain the concept of volumetric/quantitative analysis iv. define terms in volumetric analysis v. list the apparatus required for volumetric analysis vi. state and explain the necessary precaution during titration vii. outline steps in titration viii. perform acid/base and redox reactions ix. create acceptable table of titre values x. calculate unknown quantities in xi. define heat neutralization

6	**VOLUMETRIC/QUANTITATIVE ANALYSIS** **- Definition of terms: titration, trimetry, molar solution, standard solution, indicator, etc** **- Apparatus required for volumetric analysis** **- Precautions in titration** **- Steps in titration** **- Recording acceptable titre value** **- Acid-base and redox titrations** **- Determination of:** **a) concentration in g/dm and mol/dm** **b) % purity and impurity** **c) water of crystallization** **- Heat of neutralization**	At the end of the lesson, students should be able to: i. identify topics and sub-topics of the lesson ii. express the meaning of the keywords and relate such to the lesson iii. explain the concept of volumetric/quantitative analysis iv. define terms in volumetric analysis v. list the apparatus required for volumetric analysis vi. state and explain the necessary precaution during titration vii. outline steps in titration viii. perform acid/base and redox reactions ix. create acceptable table of titre values x. calculate unknown quantities in xi. define heat neutralization
7	**MID-TERM BREAK**	
8	**HYDROCARBONS** **- Definition of hydrocarbons** **- Structure and valency of carbons** **- Classification of hydrocarbons: Aliphatic or Aromatic** **- Classification of aliphatic and aromatic hydrocarbons** **- Definition of terms: homologous series, functional groups, isomerism**	At the end of the lesson, students should be able to: i. highlight keywords from the topics and sub-topics of the lesson ii. explain the meaning of the keywords and relate such to the lesson iii. discuss the occurrence of hydrocarbons iv. explain the tetravalent nature of carbon v. define and classify hydrocarbons vi. explain and define terms vii. explain 5 characteristics of a typical homologous series

9	**SATURATED HYDROCARBONS-ALKANES** **- General formula of alkanes** **- IUPAC nomenclature of alkanes** **- Laboratory preparation of alkanes (methane)** **- Properties and uses of alkanes (methane)**	At the end of the lesson, students should be able to: i. write the general formula, IUPAC name and structural formula for the first 10 alkanes ii. draw and name isomers of alkanes from a given formula iii. describe the laboratory preparation of methane iv. explain at least two chemical properties of alkanes v. list 3 uses of alkanes and methane
10	**UN-SATURATED HYDROCARBONS** **A. ALKENES** **- General formula and functional groups of alkenes** **- IUPAC nomenclature of alkenes** **- Laboratory preparation of ethene** **- Properties and uses of alkenes (ethene)** **B. ALKYNES** **-General formula of alkenes** **- Functional groups of alkenes** **- IUPAC nomenclature of alkenes** **- Laboratory preparation of ethyne** **- Properties and uses of alkenes(ethyne)** **- Test for unsaturation**	At the end of the lesson, students should be able to: i. write the general formula for alkenes and alkynes ii. specify the functional groups in alkene and alkynes iii. write the IUPAC name and structure for the first 10 alkenes and alkynes iv. describe the laboratory preparation of ethene and ethyne v. list 3 physical properties of ethene and ethyne vi. explain at least 3 chemical reactions of ethene and ethyne viii. enumerate at least 4 uses of alkenes and alkynes ix. perform experiments to test for alkenes and alkynes x. perform experiments to distinguish alkenes from alkynes

11	**ALKANOLS** **- Natural sources of alkanols** **- Classification based on:** **a) number of hydroxyl group** **b) akyl group** **- Physical and chemical properties of alkanol** **- Polarity of alkanols and solubility of substances** **- Laboratory preparation of Ethanol** **- Local and industrial preparation of ethanol by fermentation** **- Uses of alkanols** **- Test for alkanols**	At the end of the lesson, students should be able to: i. discuss natural sources of alkanols ii. list 5 substances in daily life that contain alkanols iii. write the general formula for alkanols iv. classify alkanols v. describe the laboratory and industrial preparation of alkanols vi. list and explain 5 uses of alkanols vii. prepare ethanol from starchy foods viii. explain hydrogen bonding in alkanols ix. explain and rest for different classes of alkanols
12	**REVISION/ EXAMINATION**	

3

Chapter Three
Recommended Textbooks

Recommended Textbooks

The recommended Chemistry textbooks for SSS1 include but are not limited to the following:

- New Age Chemistry by Hassan M.A, Arinze E.E, et al. New Age Press Ltd SS 1-3
- Excellence in Chemistry by A.I Oluwaseun, T. Alabi. Cambridge University Press SS1-3
- Senior Secondary Chemistry Book by S.T Bajah, B.O Onwu et al. Learn Africa SS 1-3
- Model Textbook of Chemistry by Okoye E. I. Mid-Field SS1-3
- New School Chemistry by Osei Yaw Ababio. Africana First Publisher SS 1-3
- Basic Practical Chemistry by Ema Patience Adama. University Press SS 1-3

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